Validation Flowering data

Loïc Pages

2025-11-25

Initialisation

```r
rm(list=ls())
library(spaMM)
```

```
## Registered S3 methods overwritten by 'registry':
##   method               from
##   print.registry_field proxy
##   print.registry_entry proxy
```

```
## spaMM (Rousset & Ferdy, 2014, version 4.6.42.1) is loaded. See spaMM
## ('?spaMM::spaMM') for a short introduction, 'news(package='spaMM')' for news,
## and 'citation('spaMM')' for proper citation. Further infos, slides, etc. at
## https://gitlab.mbb.univ-montp2.fr/francois/spamm-ref.
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(popbio)

setwd("/media/loic/Commun/0Travail/2025 IPM Centauree ISEM/2025 Stage ISEM/Code/IPM")

#Data
IPM_data <- read.csv("newdata.csv")

MatrixDim <- 50
RecrRate <- 0.338 #recruitment rate
```

```r
#Force maximal age to 8
AgeMax <- 8
IPM_data$Age[IPM_data$Age > AgeMax] <- AgeMax

centauree_data <- IPM_data[!is.na(IPM_data$Size0Mars) & !is.na(IPM_data$Age),]

spaMM.options(separation_max=70)
```

Kernal matrices IPM

```r
load("IPMKernalPopYear_AIC")
MatrixIPM <-  IPMKernalPopYear_AIC
load("IPMKernal_AIC")
```

Vital rate functions

```r
load("ModelsAIC")
```

```r
annees <- 1995:2022
populations <- c("E2","E1","Au","Po","Pe","Cr")

#Flowering Probability
#Flowering function for The survival-growth kernel
flr0 <- function(x, a, Beta=0,Pop,year) {
  fake_data <- expand.grid(
  year = year,
  Pop = Pop,
  Size0Mars = unique(x),
  Age = a)
  predi <- plogis(predict(Flowglm1, newdata = fake_data, allow.new.levels = T, type = "link") + Beta)
  meanpredi <- aggregate(predi, list(fake_data$Size0Mars), mean)
  return(meanpredi$V1)
}
```

```r
minsize <-  0.5
maxsize <-  25
maxsize1 <- 15
n.size <- MatrixDim
h <-  (maxsize - minsize) / n.size
h1 <-  (maxsize1 - minsize) / n.size
b = minsize + c(0:n.size) * h

#sizes for an individual of age a
#midpoint
ymid.a = 0.5 * (b[1:n.size] + b[2:(n.size + 1)])
#intervals
yI.a <- matrix(nrow = n.size, ncol = 2)
for (i in 1:n.size){
  yI.a[i,1] <- b[i]
  yI.a[i,2] <- b[i+1]
}
```

```
b1 = minsize + c(0:n.size) * h1
#sizes for an individual of age 1
#midpoint
ymid.1 = 0.5 * (b1[1:n.size] + b1[2:(n.size + 1)])
#intervals
yI.1 <- matrix(nrow = n.size, ncol = 2)
for (i in 1:n.size){
  yI.1[i,1] <- b1[i]
  yI.1[i,2] <- b1[i+1]
}
```

Validation du modèle en utilisant les données d'observation de plantes en fleur

Number of observed flowering plants since 1995.

```
Nf_tot <- c(494,420,604,638,343,490,631,482,815,435,613,794,210,NA,322,216,277,386,565,254,369,407,148,

Nf <- read.csv("flower_data.csv")
Nf <- type_convert(Nf,cols(
    Auzils = col_double(),
    Cruzades = col_double(),
    Enferrets.1 = col_double(),
    Enferrets.2 = col_double(),
    Peyrals = col_double(),
    Portes = col_double()
))
names(Nf) <- c("year","E2","E1","Au","Po","Pe","Cr")
Nf <- Nf %>% filter(year!=1994,year!=2025,year!=2024,year!=2023)
```

With IPM

We predict the number of flowering plants by population and year starting by the observed number of flowering plant the first year.

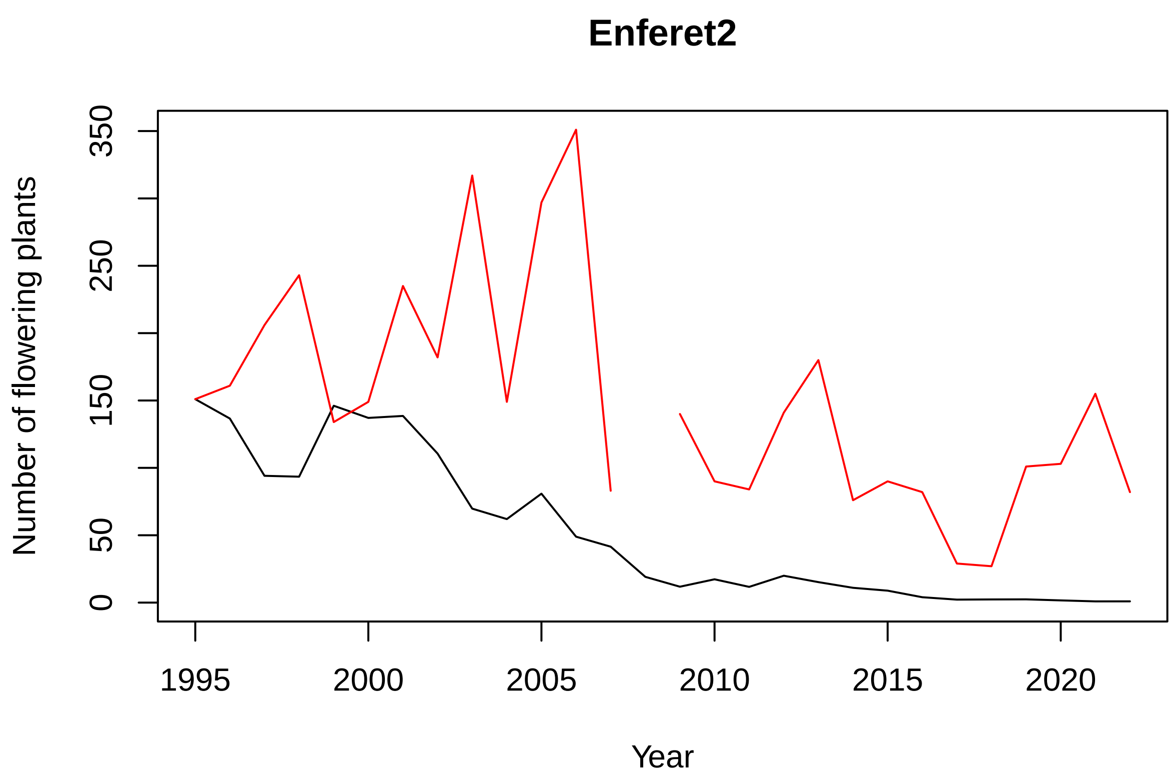
```
MatrixIPM <- IPMKernalPopYear_AIC

N <- matrix(nrow = length(populations),ncol=length(annees))
Nia <- array(data = 0, dim = c(length(populations), length(annees), 8, MatrixDim))
Nf_est <- matrix(nrow = length(populations),ncol=length(annees))

# p <- 1
for (p in 1:length(populations)) {
  # Première année
  year <- annees[1]

  # Définition de la matrice et du vecteur propre
  Pop <- populations[p]
  M <- IPMKernal_AIC
  A <- eigen.analysis(M)
```

```r
W <- function(age) {
  return(A$stable.stage[(50 * (age - 1) + 1):(50 * age)])
}

# Densité estimée d'individus fleuris
Nf_dens <- sum(W(1) * flr0(ymid.1, 1, Pop=Pop, year=year))
for (a in 2:8) {
  Nf_dens <- Nf_dens + sum(W(a) * flr0(ymid.a, a, Pop=Pop, year=year))
}
# Nombre d'individus total dans la population Pop la première année, utilisant les données observées
N[p, 1] <- Nf[1, p + 1] / Nf_dens
# Nombre d'individus dans chaque classe de taille et d'age
for (a in 1:8) {
  for (i in 1:MatrixDim) {
    Nia[p, 1, a, ] <- N[p, 1] * W(a)
  }
}

Nf_est[p, 1] <- Nf[1,p+1]

# Itération sur les années suivantes
for (yr in 2:length(annees)) {
  year <- annees[yr]
  # Définition de la matrice et du vecteur propre
  Pop <- populations[p]
  M <- MatrixIPM[[(p - 1) * length(annees) + yr]]
  A <- eigen.analysis(M)
  W <- function(age) {
    return(A$stable.stage[(50 * (age - 1) + 1):(50 * age)])
  }
  Fec <- function(a) {
    #a age de plante mère
    return(M[1:50, ((a - 1) * 50 + 1):(a * 50)])
  }
  P <- function(a) {
    #a age de plante à t
    if (a == 8) {
      return(M[((50*(a-1))+1):(50*a), (50*(a-1)+1):(50*a)])
      }
    return(M[((50*a)+1):(50*(a+1)), (50*(a-1)+1):(50*a)])}

  # Calcul du nombre d'individus dans chaque classe de taille et d'age en utilisant le kernel
  for (a in 1:8) { # Nouvelles plantules produites
    Nia[p, yr, 1, ] <- Nia[p, yr, 1, ] + Fec(a) %*% Nia[p, yr - 1, a, ]
  }
  for (a in 2:8) { # Croissance-survie des plantes
    Nia[p, yr, a, ] <- P(a) %*% Nia[p, yr - 1, a - 1, ]
  }
  Nia[p, yr, 8, ] <- Nia[p, yr, 8, ] + P(8) %*% Nia[p, yr - 1, 8, ]

  #Calcul du nombre de plantes en fleurs
  Nf_est[p, yr] <- sum(Nia[p, yr, 1, ] * flr0(ymid.1, 1, Pop=Pop, year=year))
  for (a in 2:8) {
```

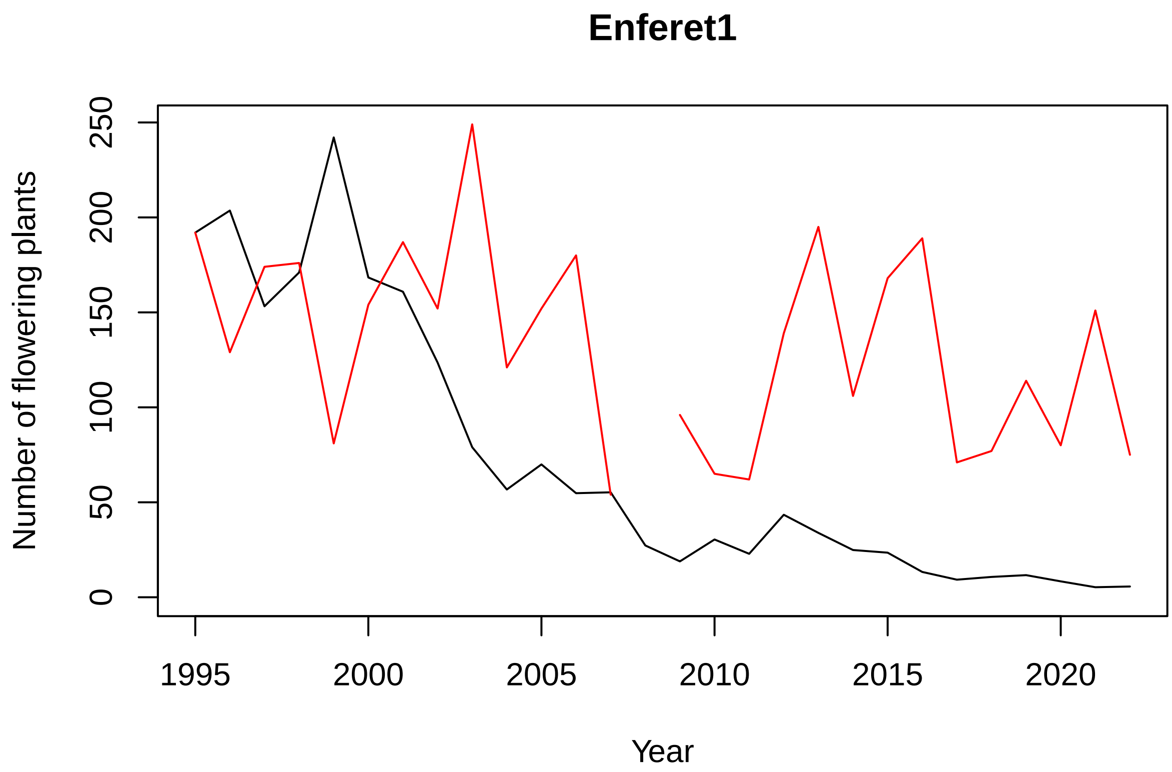
```
      Nf_est[p, yr] <- Nf_est[p, yr] + sum(Nia[p, yr, a, ] * flr0(ymid.a, a, Pop=Pop, year=year))
    }
  }
}
#Nf_est <- round(Nf_est)
```

Estimated number of flowering plants by years and population.

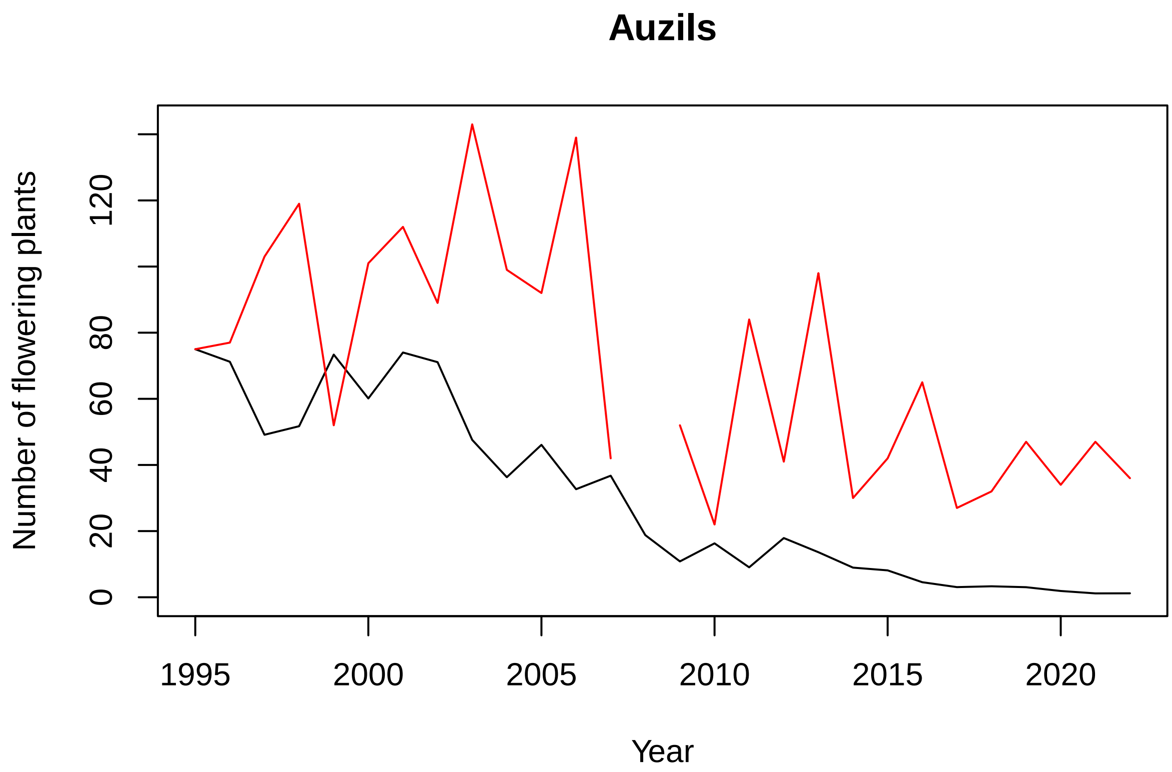
```
plot(y=Nf_est[1,], x=annees, type="l", ylim=c(0,max(c(Nf_est[1,],Nf$E2),na.rm=TRUE)), xlab="Year", ylab=
```



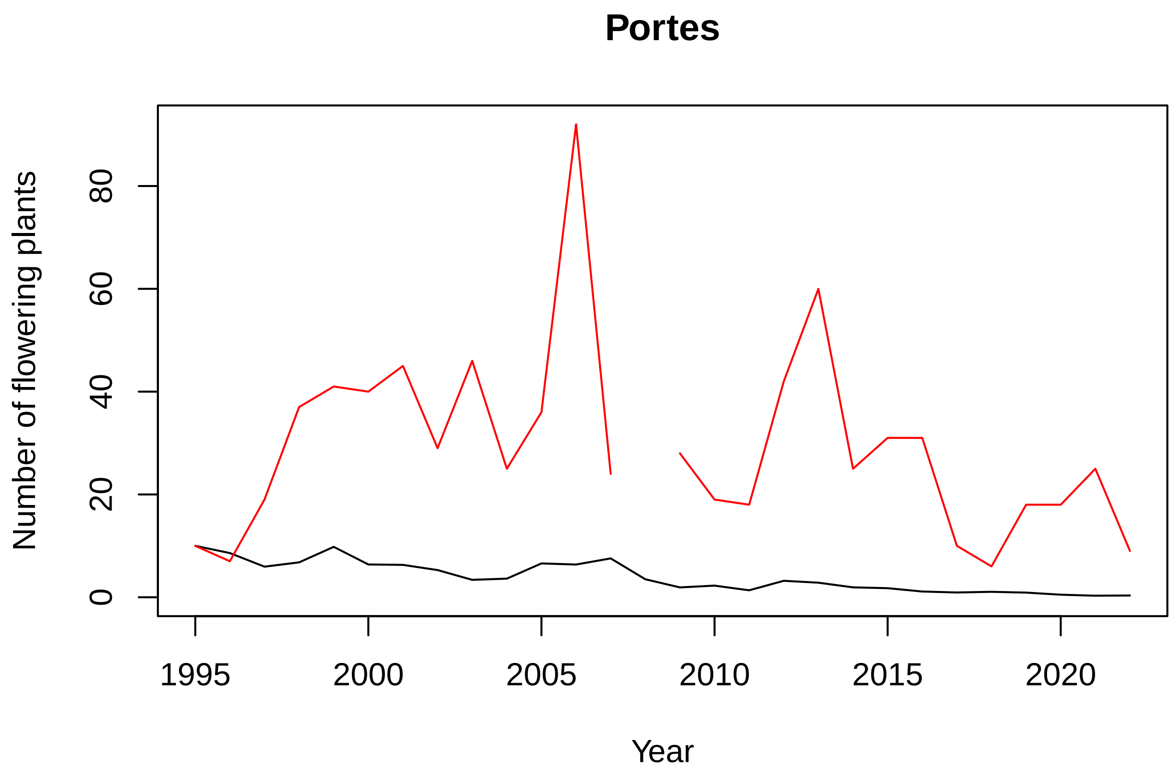

```
plot(y=Nf_est[2,], x=annees,type="l",ylim=c(0,max(c(Nf_est[2,],Nf$E1),na.rm=TRUE)), xlab="Year", ylab="
```



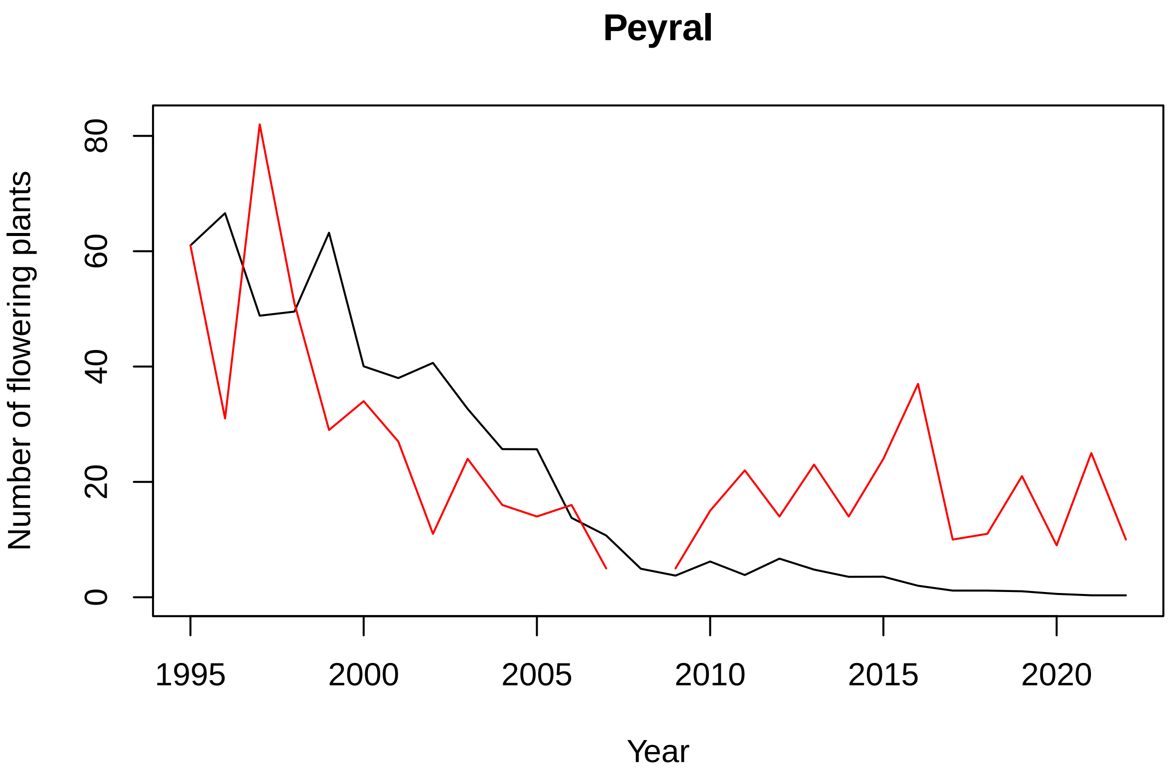

```
plot(y=Nf_est[3,], x=annees,type="l",ylim=c(0,max(c(Nf_est[3,],Nf$Au),na.rm=TRUE)), xlab="Year", ylab="N
```



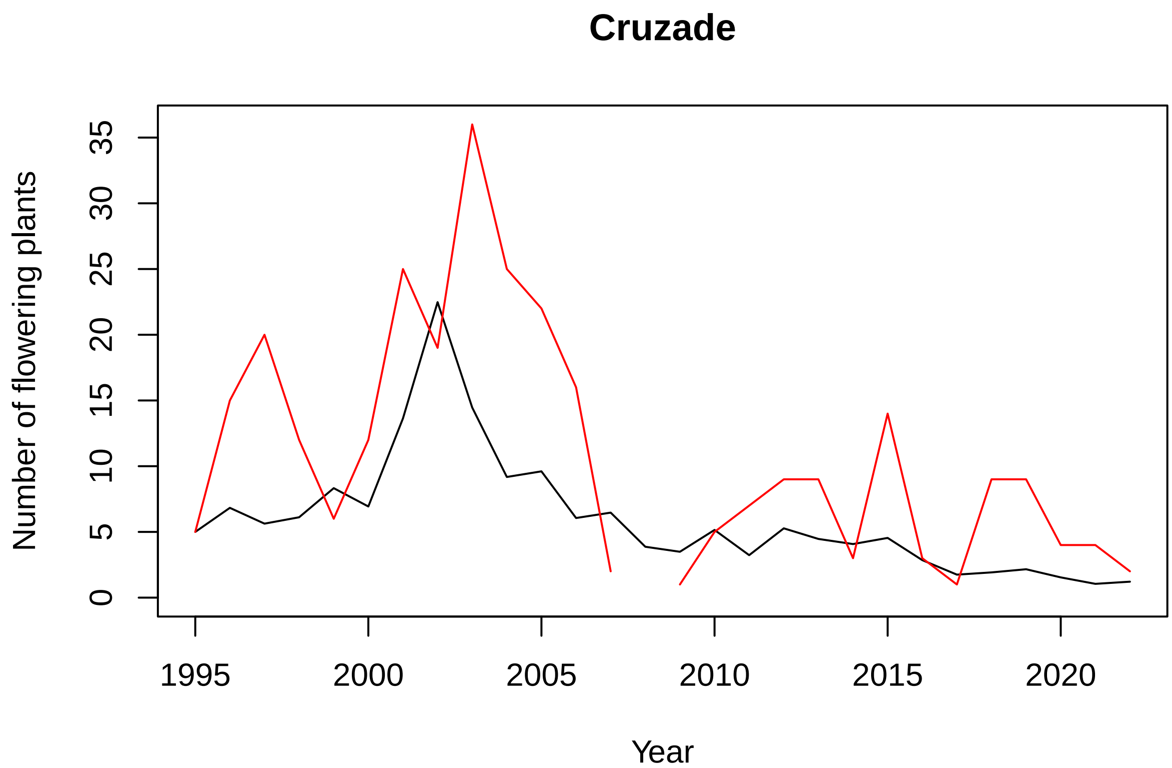

```
plot(y=Nf_est[4,], x=annees,type="l",ylim=c(0,max(c(Nf_est[4,],Nf$Po),na.rm=TRUE)), xlab="Year", ylab="
```




```
plot(y=Nf_est[5,], x=annees,type="l",ylim=c(0,max(c(Nf_est[5,],Nf$Pe),na.rm=TRUE)), xlab="Year", ylab="
```



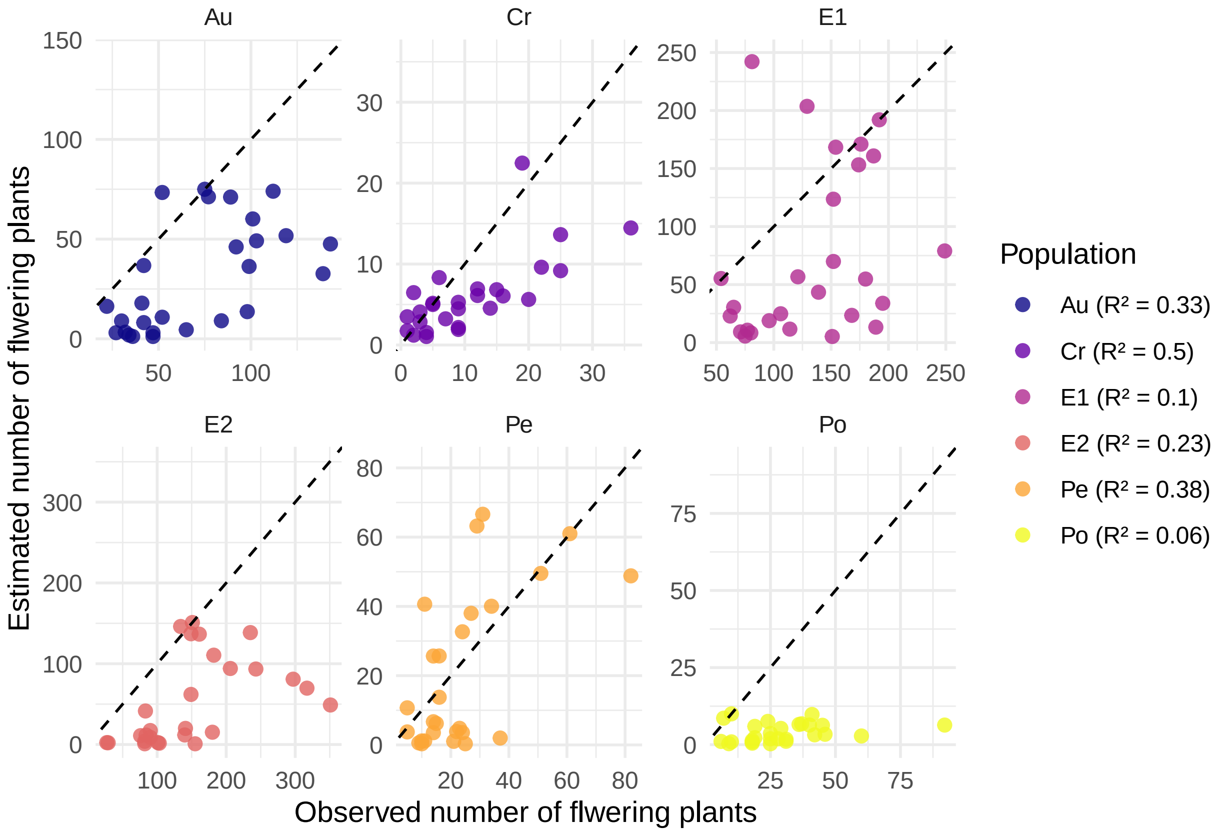

```
plot(y=Nf_est[6,], x=annees,type="l",ylim=c(0,max(c(Nf_est[6,],Nf$Cr),na.rm=TRUE)), xlab="Year", ylab="N
```



Estimated number of flowering plants by years and population compared to observed number

```
Nb_flower_IPM <- expand_grid(
  Pop = populations,
  year = annees)
Nb_flower_IPM$estimated_nb_flw <- as.vector(t(Nf_est))
Nb_flower_IPM$observed_nb_flw[Nb_flower_IPM$Pop=="E2"] <- Nf$E2
Nb_flower_IPM$observed_nb_flw[Nb_flower_IPM$Pop=="E1"] <- Nf$E1
Nb_flower_IPM$observed_nb_flw[Nb_flower_IPM$Pop=="Au"] <- Nf$Au
Nb_flower_IPM$observed_nb_flw[Nb_flower_IPM$Pop=="Po"] <- Nf$Po
Nb_flower_IPM$observed_nb_flw[Nb_flower_IPM$Pop=="Pe"] <- Nf$Pe
Nb_flower_IPM$observed_nb_flw[Nb_flower_IPM$Pop=="Cr"] <- Nf$Cr
Nb_flower_IPM <- Nb_flower_IPM %>%
  group_by(Pop) %>%
  mutate(lim = max(observed_nb_flw,estimated_nb_flw,na.rm=TRUE))

r2_data <- Nb_flower_IPM %>%
  group_by(Pop) %>%
  summarise(
    r2 = summary(lm(estimated_nb_flw ~ observed_nb_flw))$r.squared,
    biais = mean(estimated_nb_flw - observed_nb_flw, na.rm = TRUE),
    biais_relatif = 100 * biais / mean(observed_nb_flw, na.rm = TRUE))

labels_r2 <- paste0(r2_data$Pop, " (R² = ", round(r2_data$r2, 2), ")")
names(labels_r2) <- r2_data$Pop
```

```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



With MPM

```
anneesMPM <- annees[-c(27,28)]
```

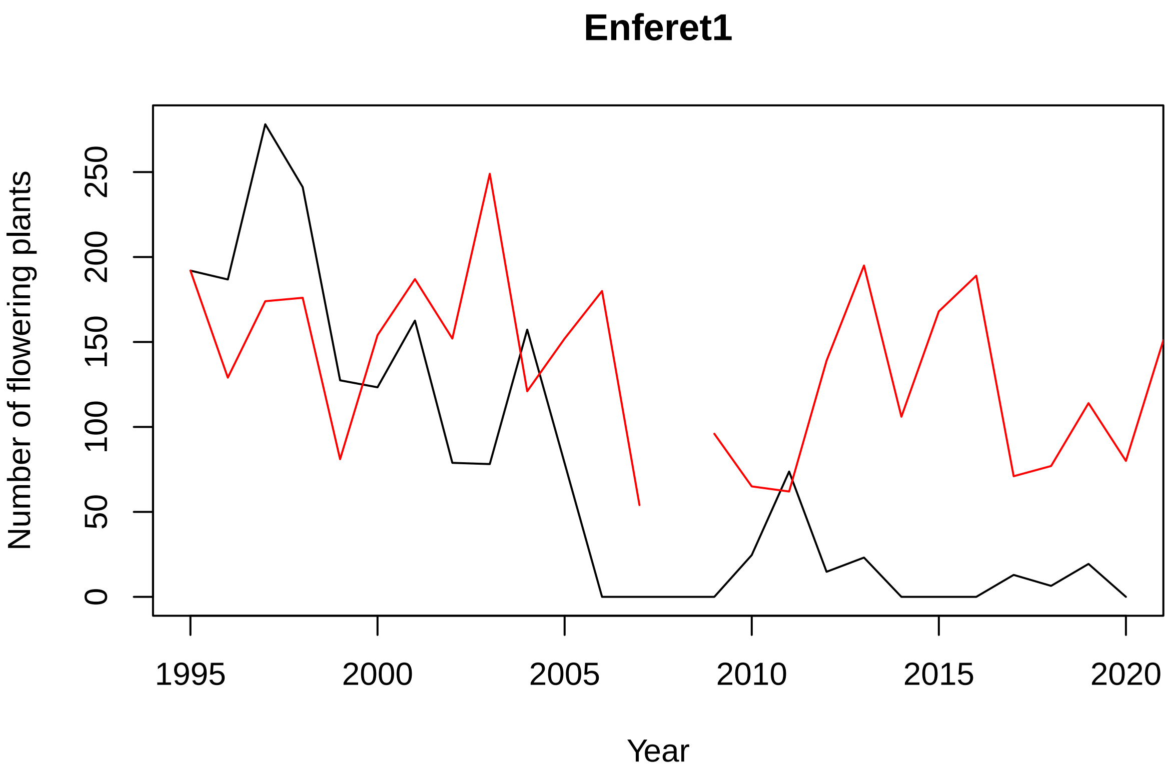
Kernal matrices MPM

```
load("/media/loic/Commun/0Travail/2025 IPM Centauree ISEM/2025 Stage ISEM/Bibliographie/Code AsmaEric/dy
matricesE2 <- matrices[-1]
load("/media/loic/Commun/0Travail/2025 IPM Centauree ISEM/2025 Stage ISEM/Bibliographie/Code AsmaEric/dy
matricesE1 <- matrices[-1]
load("/media/loic/Commun/0Travail/2025 IPM Centauree ISEM/2025 Stage ISEM/Bibliographie/Code AsmaEric/dy
matricesAu <- matrices[-1]
load("/media/loic/Commun/0Travail/2025 IPM Centauree ISEM/2025 Stage ISEM/Bibliographie/Code AsmaEric/dy
matricesPo <- matrices[-1]
load("/media/loic/Commun/0Travail/2025 IPM Centauree ISEM/2025 Stage ISEM/Bibliographie/Code AsmaEric/dy
matricesPe <- matrices[-1]
load("/media/loic/Commun/0Travail/2025 IPM Centauree ISEM/2025 Stage ISEM/Bibliographie/Code AsmaEric/dy
matricesCr <- matrices[-1]
rm(matrices)
load("/media/loic/Commun/0Travail/2025 IPM Centauree ISEM/2025 Stage ISEM/Bibliographie/Script pour vér
```

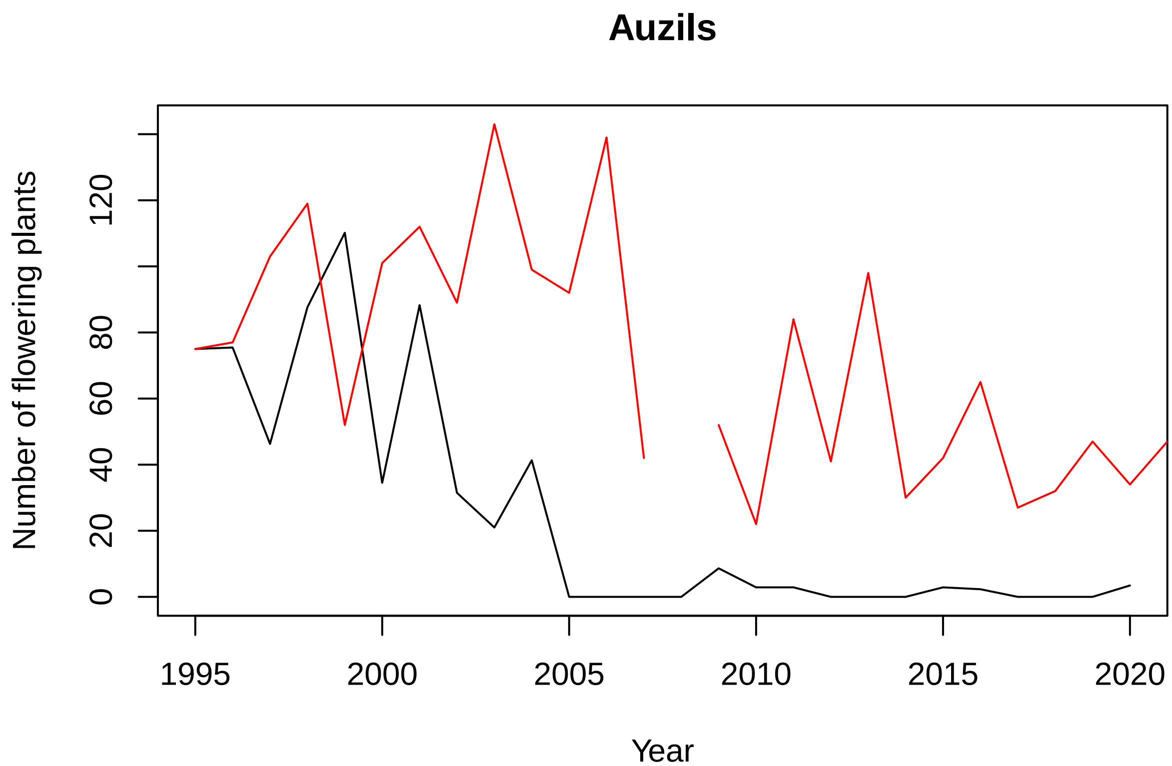
```
Nf_est_M = matrix(nrow = length(populations),ncol=length(matricesAu))
N <- NA

for (p in 1:length(populations)) {
  # Définition de la matrice et du vecteur propre
  Pop <- populations[p]
  if (Pop=="E2") {matrices <- matricesE2}
  if (Pop=="E1") {matrices <- matricesE1}
  if (Pop=="Au") {matrices <- matricesAu}
  if (Pop=="Po") {matrices <- matricesPo}
  if (Pop=="Pe") {matrices <- matricesPe}
  if (Pop=="Cr") {matrices <- matricesCr}

  # Définition de la matrice et du vecteur propre
    M <- A
    ev <- eigen(M)
    W <- abs(Re(ev$vectors[,1]))
    W <- W/sum(W)
    N <- round(W*Nf[1,p+1]/W[3])
    Nf_est_M[p,1] <- Nf[1,p+1]

  for (i in 2:26){
    M <- matrices[[i]]
    ev <- eigen(M)
    W <- abs(Re(ev$vectors[,1]))
    W <- W/sum(W)
    # if(is.nan(N[1]==TRUE)){N <- W*Nf_est_M[p,i-1]/W[3]}
    N <- M%*%N
    Nf_est_M[p,i] <- N[3]
  }
}
```

```
plot(y=Nf_est_M[1,], x=anneesMPM, type="l", ylim=c(0,max(c(Nf_est_M[1,],Nf$E2),na.rm=TRUE)), xlab="Year
```

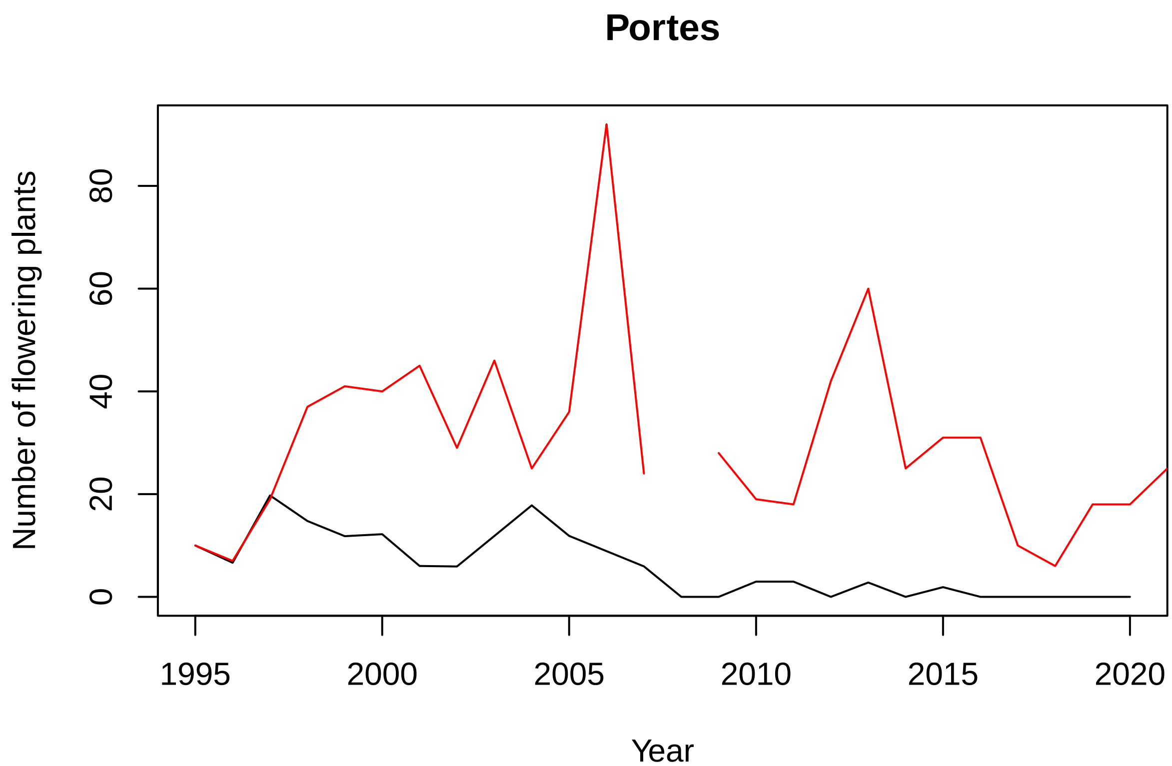

```
plot(y=Nf_est_M[2,], x=anneesMPM,type="l",ylim=c(0,max(c(Nf_est_M[2,],Nf$E1),na.rm=TRUE)), xlab="Year",
```



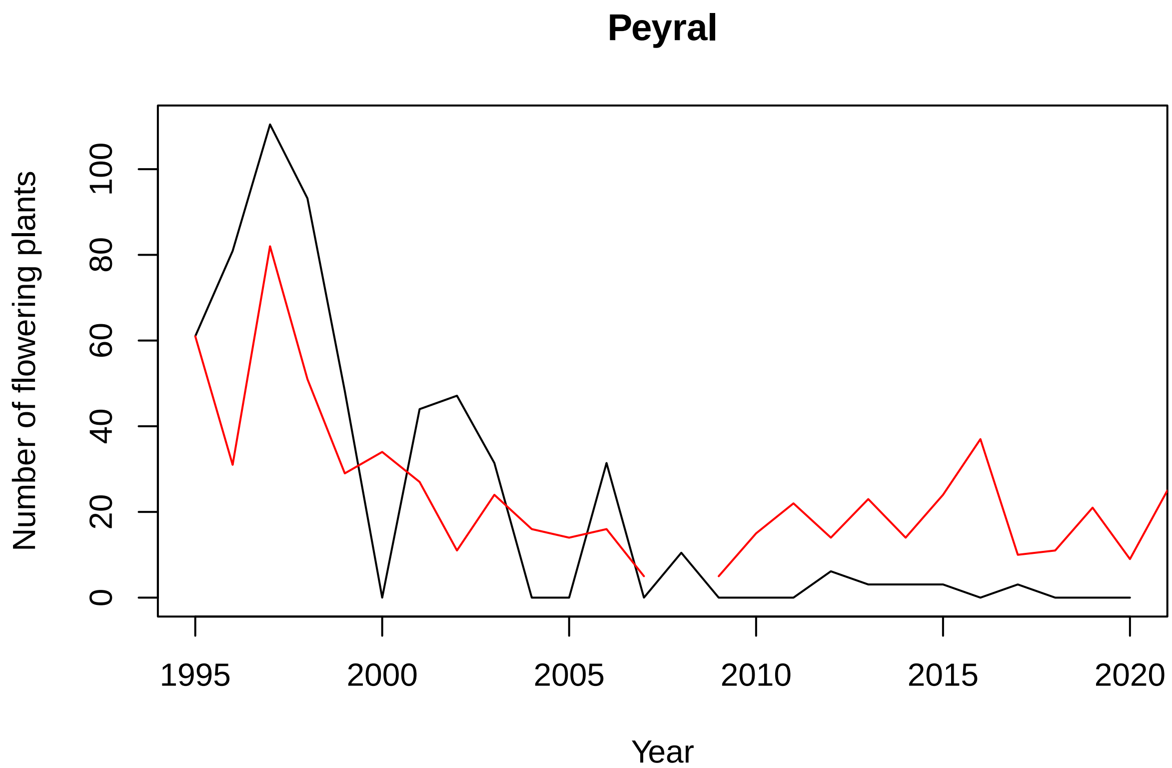

```
plot(y=Nf_est_M[3,], x=anneesMPM,type="l",ylim=c(0,max(c(Nf_est_M[3,],Nf$Au),na.rm=TRUE)), xlab="Year",
```



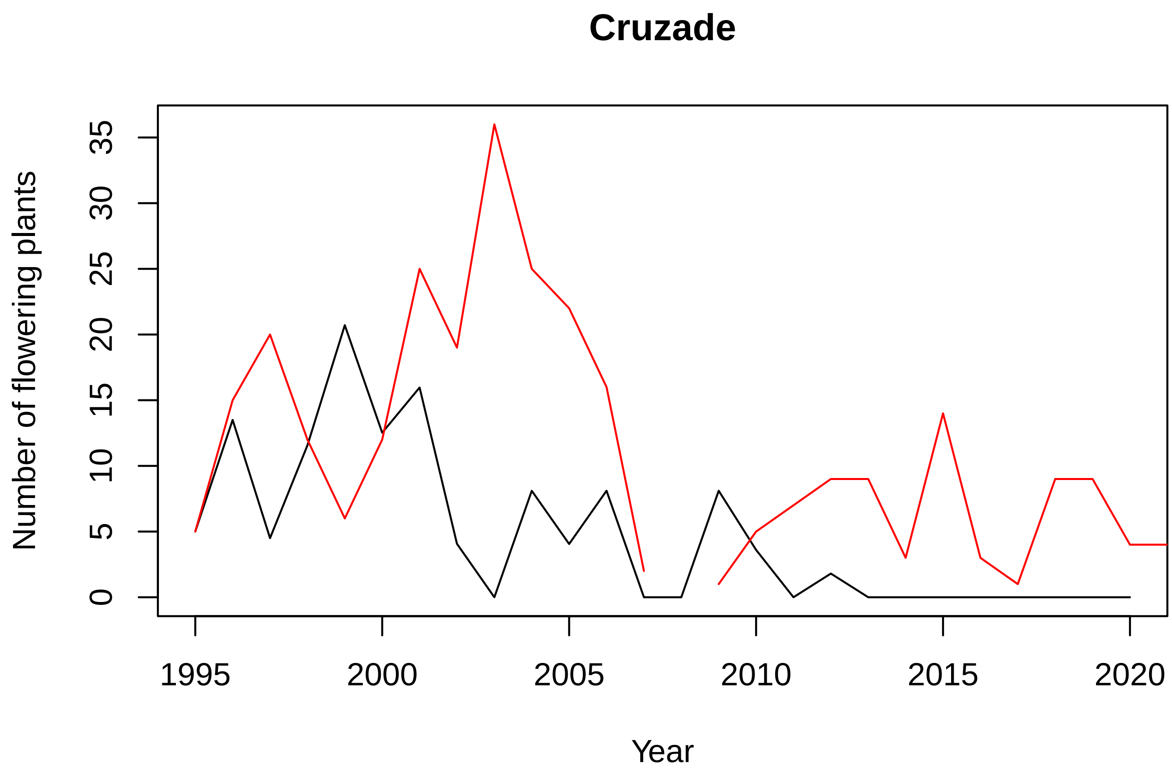

```
plot(y=Nf_est_M[4,], x=anneesMPM,type="l",ylim=c(0,max(c(Nf_est_M[4,],Nf$Po),na.rm=TRUE)), xlab="Year",
```




```
plot(y=Nf_est_M[5,], x=anneesMPM,type="l",ylim=c(0,max(c(Nf_est_M[5,],Nf$Pe),na.rm=TRUE)), xlab="Year",
```

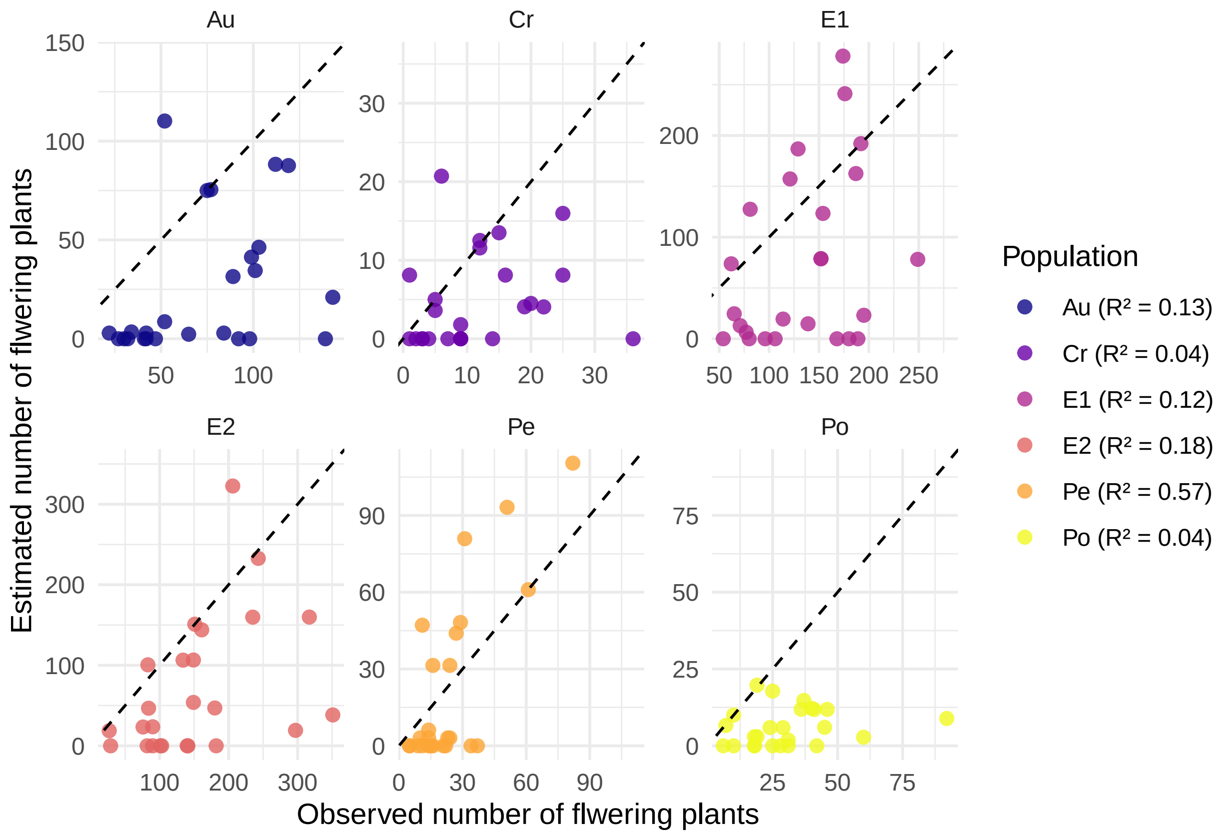



```
plot(y=Nf_est_M[6,], x=anneesMPM,type="l",ylim=c(0,max(c(Nf_est_M[6,],Nf$Cr),na.rm=TRUE)), xlab="Year",
```

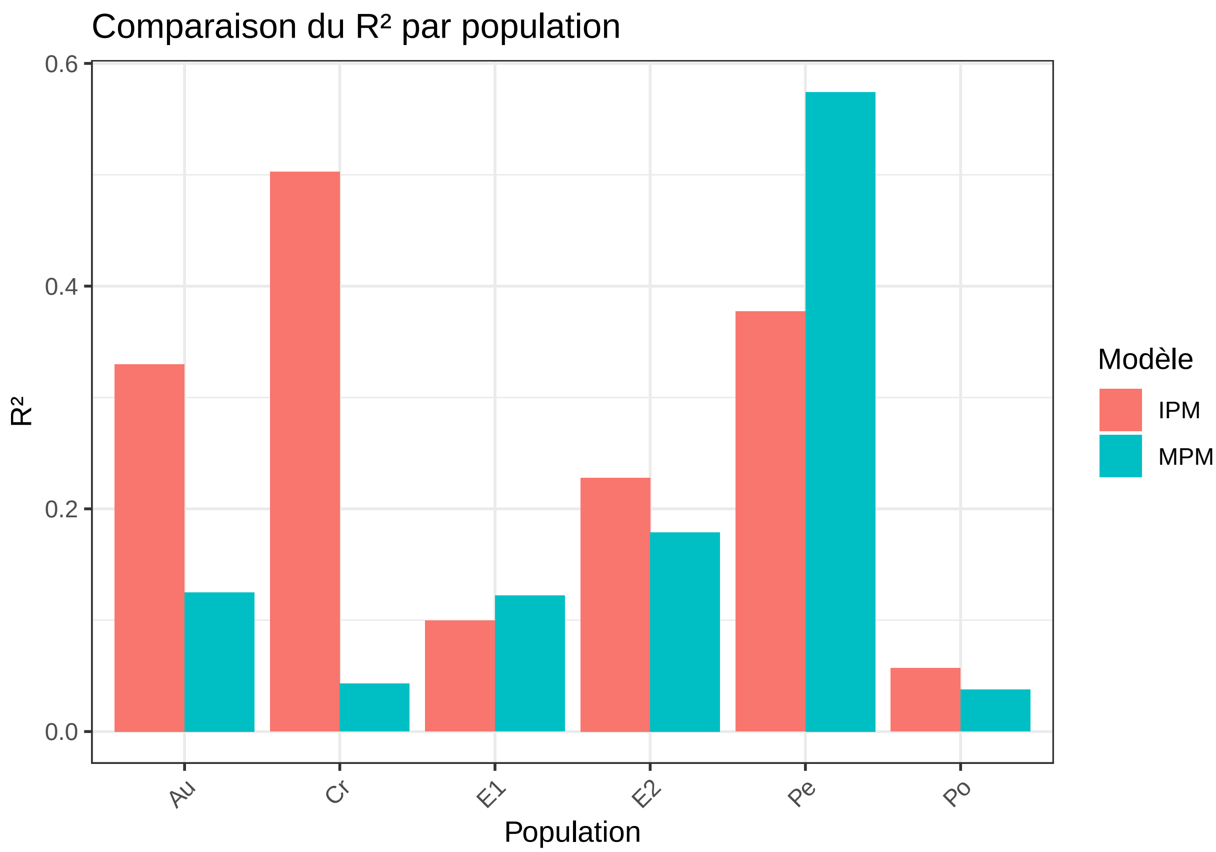


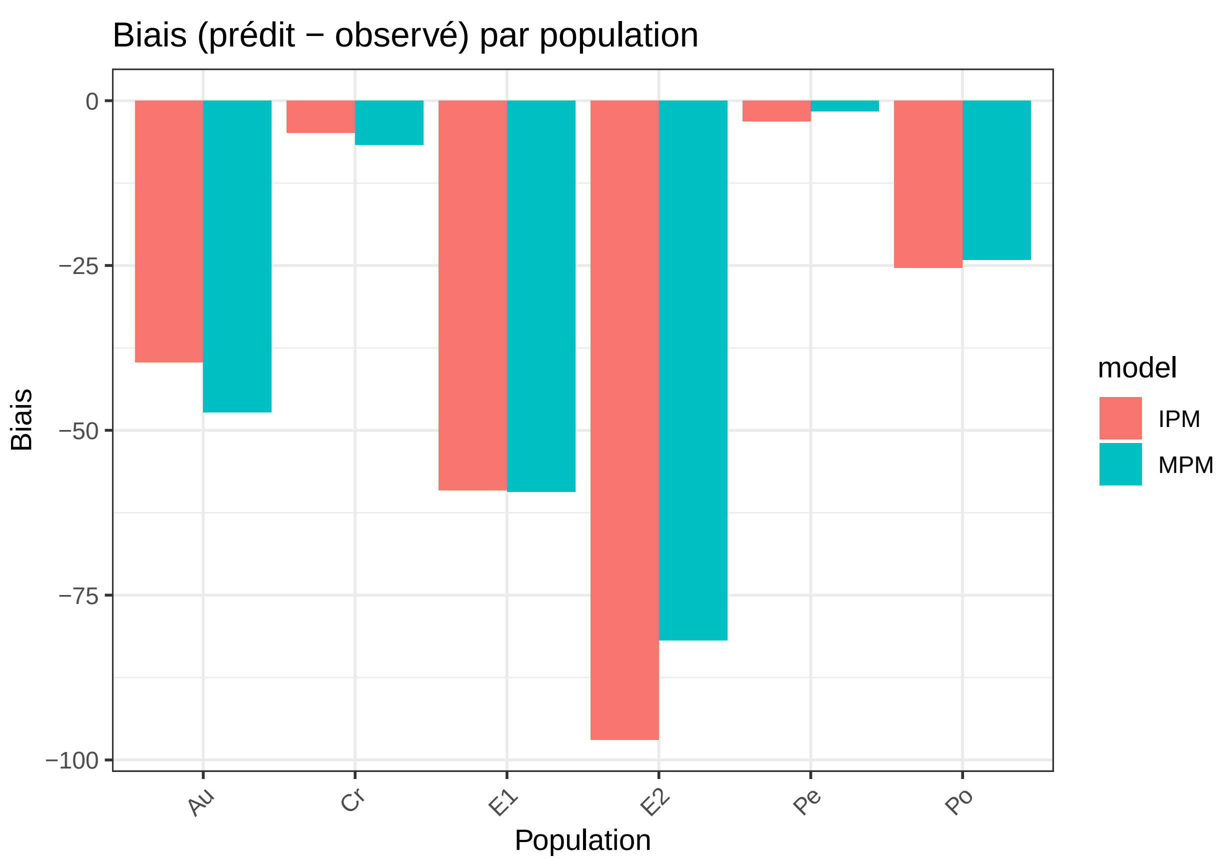

```
Nb_flower_MPM <- expand_grid(
  Pop = populations,
  year = annees[-c(27,28)])
Nb_flower_MPM$estimated_nb_flw <- as.vector(t(Nf_est_M))
Nb_flower_MPM$observed_nb_flw[Nb_flower_MPM$Pop=="E2"] <- Nf$E2
Nb_flower_MPM$observed_nb_flw[Nb_flower_MPM$Pop=="E1"] <- Nf$E1
Nb_flower_MPM$observed_nb_flw[Nb_flower_MPM$Pop=="Au"] <- Nf$Au
Nb_flower_MPM$observed_nb_flw[Nb_flower_MPM$Pop=="Po"] <- Nf$Po
Nb_flower_MPM$observed_nb_flw[Nb_flower_MPM$Pop=="Pe"] <- Nf$Pe
Nb_flower_MPM$observed_nb_flw[Nb_flower_MPM$Pop=="Cr"] <- Nf$Cr
Nb_flower_MPM <- Nb_flower_MPM %>%
  group_by(Pop) %>%
  mutate(lim = max(observed_nb_flw,estimated_nb_flw,na.rm=TRUE))

r2_data_M <- Nb_flower_MPM %>%
  group_by(Pop) %>%
  summarise(
    r2 = summary(lm(estimated_nb_flw ~ observed_nb_flw))$r.squared,
    biais = mean(estimated_nb_flw - observed_nb_flw, na.rm = TRUE),
    biais_relatif = 100 * biais / mean(observed_nb_flw, na.rm = TRUE))

labels_r2_M <- paste0(r2_data_M$Pop, " (R² = ", round(r2_data_M$r2, 2), ")")
names(labels_r2_M) <- r2_data_M$Pop
```

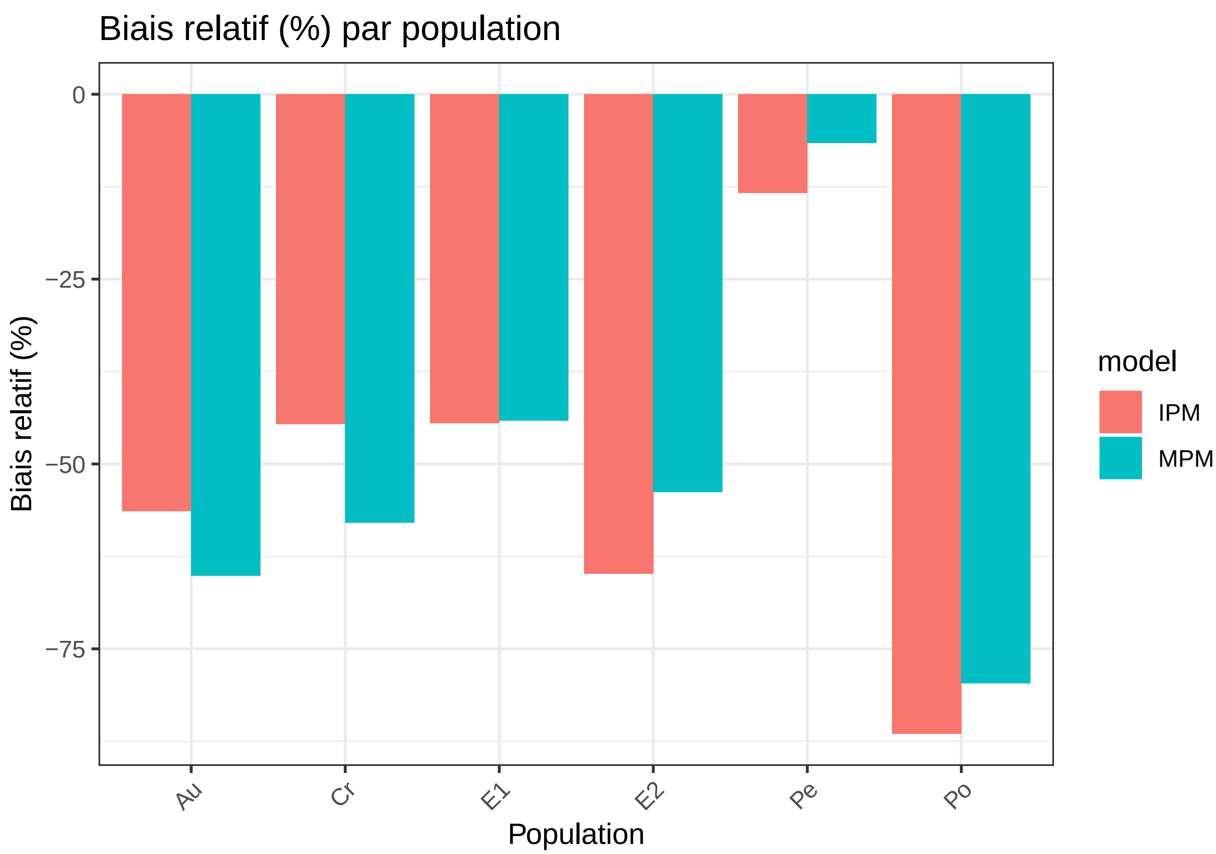
```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```


Au
Cr
E1
E2
Pe
Po
Estimated number of flwering plants
Observed number of flwering plants
Population
Au (R² = 0.13)
Cr (R² = 0.04)
E1 (R² = 0.12)
E2 (R² = 0.18)
Pe (R² = 0.57)
Po (R² = 0.04)

Comparaison r2 et biais







Biais relatif (%) par population
0
−25
−50
−75
Biais relatif (%)
Au
Cr
E1
E2
Pe
Po
Population
model
IPM
MPM